AST4762C Extra credit

Aramis Birkhimer

4a) $f(x) = \frac{A}{2} = Ae^{-x^2/2\sigma^2}$, $\ln(1/2) = \frac{-x^2}{2\sigma^2}$

$\ln(2) = x^2/2\sigma^2$

$\sqrt{2\sigma^2\ln(2)} = x = \sigma\sqrt{2\ln(2)}$

$FWHM = 2x_{1/2}$

$\boxed{FWHM = 2\sqrt{2\ln(2)}\,\sigma \approx 2.355\sigma}$

4b) $y = ax^k$, $\log(y) = k\log(x) + \log(a)$

Let $y = \log(y)$, $x = \log(x)$, $m, b = k, \log(a)$

↳ $y = mx + b$ ✓

$Slope = m = k = \frac{\log(y) - \log(a)}{\log(x)}$

$\log(y(x_2)) = m\log(x_2) + \log(a)$ ← b

$-\ \log(y(x_1)) = m\log(x_1) + b$

$\log(y(x_2)/y(x_1)) = m\log(x_2/x_1)$

$\boxed{m = \frac{\log_{10}(y(x_2)/y(x_1))}{\log_{10}(x_2/x_1)}}$